

Mr. Wycliff Gakenge

Geospatial Engineer | GIS Developer | Python Developer

Nairobi, Kenya

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GitHub: <https://github.com/Gakenge0>

Portfolio: <https://gakenge0.github.io/Gakenge.github.io/>

CAREER

Geospatial Engineer with a strong foundation in GIS, remote sensing, surveying, and geospatial software development. Experienced in designing spatial workflows, managing geospatial databases, and developing data-driven solutions using Python, ArcGIS Pro, QGIS, SQL/PostGIS, and web technologies. Passionate about building scalable GIS applications, automating spatial processes, and transforming complex geospatial data into actionable insights. Driven by a commitment to leveraging technology to address environmental, infrastructure, and climate-related challenges through innovative geospatial solutions.

SKILLS

GIS & Geospatial Technologies

- ArcGIS Pro, QGIS
- Spatial Analysis
- Cartography & Map Design
- Geospatial Data Management
- Coordinate Reference Systems (CRS)
- GNSS/RTK Data Collection
- Topographic & Boundary Surveys
- Photogrammetry

Remote Sensing

- Google Earth Engine (JavaScript API)
- Landsat & Sentinel Imagery Processing
- Image Classification
- NDVI & Vegetation Analysis
- Raster Processing & Spatial Modelling

Programming & GIS Development

- Python
- JavaScript
- HTML5 & CSS3
- Django
- SQL
- GIS Workflow Automation
- REST API Fundamentals

Spatial Databases

- PostgreSQL
- PostGIS
- MySQL
- Spatial Data Design & Management

Data Science & Visualization

- R
- Tableau
- Python (NumPy, Pandas, Matplotlib)
- Data Cleaning & Analysis
- Interactive Data Visualization

Machine Learning

- Scikit-learn
- TensorFlow
- Supervised & Unsupervised Learning
- Geospatial Machine Learning

Professional Skills

- Analytical Problem Solving
- Technical Documentation
- Communication & Stakeholder Engagement
- Team Collaboration
- Adaptability
- Continuous Learning

TECHNICAL PROJECTS

GIS & Remote Sensing-Based Analysis of Deforestation-Driven Soil Erosion

Technologies: Google Earth Engine, ArcGIS Pro, QGIS, Python, Landsat, Sentinel-2

- Developed a comprehensive geospatial workflow to assess the impact of deforestation on soil erosion using remote sensing and GIS techniques.
- Processed multi-temporal Landsat and Sentinel-2 imagery to analyze land cover changes and vegetation dynamics.
- Applied the Revised Universal Soil Loss Equation (RUSLE) to model and map soil erosion susceptibility.
- Integrated environmental datasets, including Digital Elevation Models (DEMs), rainfall, land cover, and soil information, to support spatial modelling.
- Produced maps, analytical outputs, and visualizations to support environmental decision-making and sustainable land management.

Geospatial Database Design & Spatial Analysis

Technologies: PostgreSQL, PostGIS, QGIS, SQL

- Designed and managed spatial databases for storing, querying, and analyzing geospatial datasets.
- Developed efficient spatial queries to improve data retrieval and support GIS analysis.
- Implemented structured database workflows to enhance spatial data integrity and accessibility.

GIS Workflow Automation & Spatial Data Processing

Technologies: Python, ArcGIS Pro, QGIS

- Developed Python-based workflows to streamline repetitive GIS tasks and improve data processing efficiency.
- Automated spatial data preparation, analysis, and visualization processes.
- Applied scripting techniques to enhance productivity and ensure consistency across GIS workflows.

Personal Geospatial Portfolio Website

Technologies: HTML5, CSS3, JavaScript

- Designed and developed a responsive personal portfolio website to showcase geospatial projects, technical skills, and professional experience.
- Implemented modern UI/UX principles, responsive layouts, and interactive project presentations.

- Created a platform to demonstrate both software development capabilities and geospatial expertise.

PROFESSIONAL EXPERIENCE

Customer Care Representative

CCI Kenya Ltd (Metro by T-Mobile Campaign) | October 2025 – July 2026

- Delivered technical customer support by resolving inquiries related to mobile services, account management, billing, device activation, and network connectivity.
- Applied structured troubleshooting methodologies to diagnose and resolve customer issues while consistently achieving first-contact resolution.
- Managed high volumes of customer interactions while maintaining accuracy, professionalism, and compliance with data privacy and quality assurance standards.
- Collaborated with cross-functional teams to resolve complex customer concerns and improve service delivery.
- Strengthened communication, analytical problem-solving, conflict resolution, and time management skills within a fast-paced, performance-driven environment.

Geospatial Engineering Attaché

Ministry of Lands, Public Works, Housing and Urban Development – Nyeri County | August 2024 – September 2024

- Participated in cadastral, fixed boundary, and general boundary surveys using modern surveying techniques and GNSS equipment.
- Processed survey observations and produced accurate cadastral maps and plans using AutoCAD and Compass Solution software.
- Assisted in field data collection, verification, and quality assurance to ensure compliance with surveying standards and regulations.
- Collaborated with survey teams to support land administration, boundary determination, and geospatial data management activities.
- Developed practical experience in integrating field observations with digital mapping and spatial data processing workflows.

Geospatial Engineering Intern

University of Nairobi | July 2024 – August 2024

- Developed geospatial workflows using Google Earth Engine, ArcGIS Pro, and QGIS for remote sensing analysis and spatial data processing.
- Designed and managed spatial databases using PostgreSQL and QGIS to support efficient storage, retrieval, and analysis of geospatial data.
- Produced geospatial visualizations and analytical outputs using R to communicate complex spatial information effectively.
- Conducted topographic, engineering, levelling, and GNSS/RTK surveys while ensuring accurate field data collection and processing.
- Contributed to hydrographic survey field activities through collaboration with the Kenya Marine and Fisheries Research Institute (KMFRI), gaining exposure to specialized geospatial data acquisition techniques. diverse customer concerns under demanding service-level expectations.

PROFESSIONAL COURSES

Bachelor of Science in Geospatial Engineering

University of Nairobi | Graduated: September 2025

Relevant Coursework:

- Geographic Information Systems (GIS)
- Remote Sensing

- Photogrammetry
- Geodesy
- Cartography
- Spatial Databases
- Survey Computations
- Engineering Surveying
- Hydrographic Surveying
- Land Information Systems
- Geographic Information Science

PROFESSIONAL COURSES

- **Python for Data Science** | Coursera | May 2022- August 2022 | <https://coursera.org/share/8f0110417d3aab1c3c8d7c1d68b3a7a6>
- **Spatial Data Science** | Younsei University_ Coursera | August 2023 – February 2024 | <https://coursera.org/share/b3d93845c0412adb8590449e30742891>
- **Web Development** (JavaScript and Python) | Harvard CS50
- **Use of CORS network** | Measurements Systems LTD | November 2023
- **Certified Space Professional** | Teaching Science and Technology Inc. | July 2024 -August 2024
<https://www.credly.com/badges/0d4533ee-fe4c-4c85-b2eb-6b9ff2f6718e>
- **Global Mentorship Initiative (GMI) Graduate**
Global Mentorship Initiative | March 2024 – February 2025
- **Software Development with Python and Database Design** | Power Learn Project | July 2025 – November 2025

LANGUAGES

- English (Fluent)
- Swahili (Fluent)
- French (Basic)

HOBBIES

- Poetry
- Football
- Travelling

REFERENCES

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